

Name: Maths Class:

SYDNEY TECHNICAL HIGH SCHOOL



Year 12 Mathematics Extension 1

HSC Course

Assessment 3

June, 2015

Time allowed: 90 minutes

General Instructions:

- Marks for each question are indicated on the question.
- Approved calculators may be used
- All necessary working should be shown
- Full marks may not be awarded for careless work or illegible writing
- ***Begin each question on a new page***
- Write using black or blue pen
- All answers are to be in the writing booklet provided
- A set of Standard Integrals is provided at the rear of this Question Booklet, and may be removed at any time.

Section 1 Multiple Choice
Questions 1-5
5 Marks

Section II Questions 6-11
55 Marks

SECTION 1

Attempt questions 1-5

5 Marks

Use multiple choice answer sheet

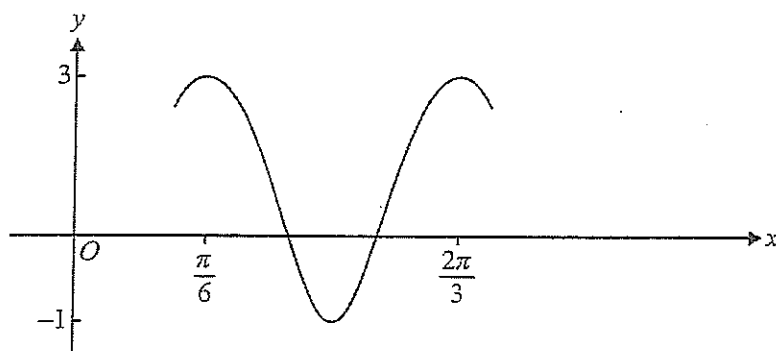
1.

The inverse function of $g(x)$, where $g(x) = \sqrt{2x-4}$ is

- (A) $g^{-1}(x) = \frac{x^2+4}{2}$
- (B) $g^{-1}(x) = (2x-4)^2$
- (C) $g^{-1}(x) = \sqrt{\frac{x}{2}} + 4$
- (D) $g^{-1}(x) = \frac{x^2-4}{2}$

2.

The graph below could have the equation



- (A) $y = 2\cos\left(x + \frac{\pi}{6}\right) + 1$
- (B) $y = 2\cos 2\left(x + \frac{\pi}{6}\right) + 1$
- (C) $y = 2\cos 4\left(x - \frac{\pi}{6}\right) + 1$
- (D) $y = 2\cos 4\left(x + \frac{2\pi}{3}\right) + 1$

3.

The domain and range of the function $f(x)$, where $f(x) = 3\sin^{-1}(4x-1)$ are respectively.

- (A) $0 \leq x \leq \frac{1}{2}$ and $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$
- (B) $-\frac{1}{2} \leq x \leq 0$ and $-\frac{\pi}{2} \leq y \leq \frac{5\pi}{2}$
- (C) $0 \leq x \leq \frac{1}{2}$ and $-\frac{\pi}{2} \leq y \leq \frac{5\pi}{2}$
- (D) $-\frac{1}{2} \leq x \leq 0$ and $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$

4.

If the substitution $u = x^2 - 1$ is used then the definite integral $\int_0^2 \frac{x}{\sqrt{x^2 - 1}} dx$ can be simplified to

(A) $\frac{1}{2} \int_{-1}^3 u^{-\frac{1}{2}} du$

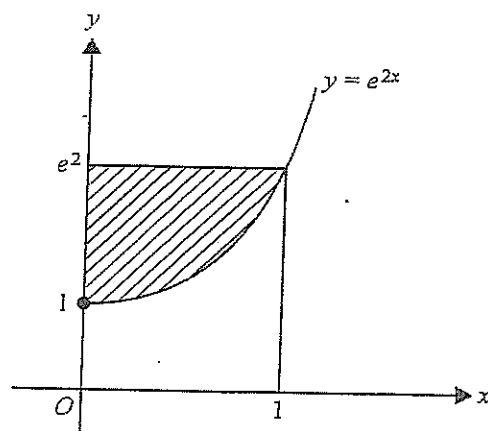
(B) $2 \int_{-1}^3 u^{-\frac{1}{2}} du$

(C) $\frac{1}{2} \int_0^2 u^{-\frac{1}{2}} du$

(D) $2 \int_0^2 u^{-\frac{1}{2}} du$

5.

To find the area of the shaded region in the diagram below, four different students proposed the following calculations.



Student 1: $\int_0^1 e^{2x} dx$

Student 2: $e^2 - \int_0^1 e^{2x} dx$

Student 3: $\int_1^{e^2} e^{2y} dy$

Student 4: $\int_1^{e^2} \frac{\log_e y}{2} dy$

Which of the following is correct?

(A) Student 2 only.

(C) Students 2 and 4 only.

(B) Students 2 and 3 only.

(D) Students 1 and 4 only.

SECTION II

Question 6 (9 Marks)

Mark

a) Differentiate

i) $e^{\sin x}$

1

ii) $\ln(\cos x)$

1

iii) $\sin^{-1} \sqrt{x}$

2

b) Find the exact values of

i) $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$

1

ii) $\tan^{-1}\left(2\cos\frac{5\pi}{6}\right)$

2

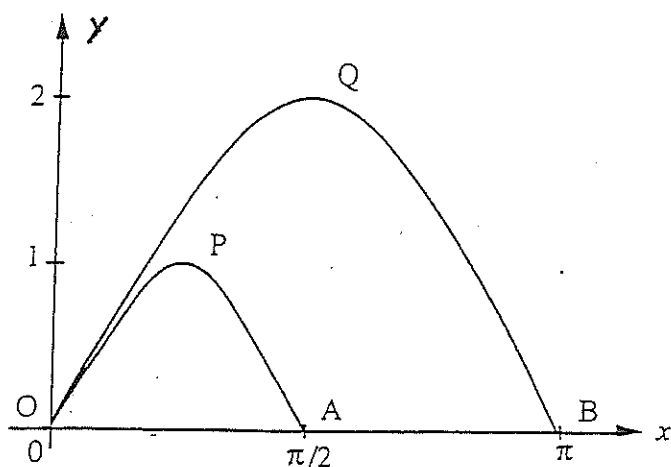
c) Evaluate $\int_0^1 \frac{1}{\sqrt{4-x^2}} dx$

2

Question 7 (9 Marks) (Start a new page)

a)

3



The diagram shows portions of the graphs of

$$y = 2\sin x \text{ and } y = \sin 2x$$

Calculate the area of the region bounded by the arc OPA, the arc OQB and the interval AB.

			Mark
b)	i)	Find $\frac{d}{dx}(x \ln x)$	2
	ii)	Hence prove $\int_e^{e^2} \frac{1+\ln x}{x \ln x} dx = 1 + \ln 2$	2
c)	i)	Write $\cos 2x$ in terms of $\sin^2 x$	1
	ii)	Hence or otherwise find	1

$$\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$$

Question 8 (10 Marks) (Start a new page)

a)		If $y = a \cdot 10^{bx}$ make x the subject	2
b)		Find the general solution of $\sin x = -\frac{1}{2}$	2
c)		The gradient function of a curve is given by $\frac{dy}{dx} = \frac{2}{x+1}$. If the curve passes through the point $(0, 1)$, find the equation of the curve.	2
d)	i)	Express $\sin^2 x \cos^2 x$ in terms of $\sin 2x$	1
	ii)	Hence find $\int \sin^2 x \cos^2 x dx$	3

Question 9 (9 Marks) (Start a new page)

a)	i)	Sketch $g(x) = (x - 2)^2 - 3$ showing and labelling the vertex and y intercept.	1
	ii)	What is the largest domain containing $x = 0$ for which $g(x)$ has an inverse?	1
	iii)	Find the inverse function $g^{-1}(x)$ and sketch it on your diagram showing where it cuts the x axis.	2

Label your curve clearly

		Mark
b)	i) Differentiate $y = \cos^{-1} x + \sin^{-1} x$	1
	ii) Hence sketch $y = \cos^{-1} x + \sin^{-1} x$ (Label both axes and show a suitable scale)	2
c)	Find $\int \sec^2 x \cdot \tan x \, dx$ by using the substitution $u = \tan x$ or otherwise.	2

Question 10 (8 Marks) (Start a new page)

a)	i) Sketch $y = 1 - \frac{2}{x}$ (do not use calculus) and indicate on your sketch any asymptotes and where the curve cuts the x axis.	2
	ii) The region bounded by the curve and the x axis from $x = 1$ to $x = 2$ is rotated around the x axis Show the volume generated is $\pi (3 - 4 \ln 2)$ units ³	2
b)	i) Find the co-ordinates of the stationary point on the graph of $y = \frac{e^x}{x^2 + 1}$ and prove it is neither a maximum nor a minimum.	2
	ii) Sketch $y = \frac{e^x}{x^2 + 1}$ showing the stationary point and where the curve cuts the y axis and any asymptotes.	2

Question 11 (10 Marks) (Start a new page)

a)	i) Prove $\frac{1}{x-2} - \frac{1}{x+2} = \frac{4}{x^2-4}$	1
	ii) Hence find $\int_3^6 \frac{1}{x^2-4} \, dx$ in exact form	2
b)	i) If $\tan^{-1} x = \alpha$ and $\tan^{-1} y = \beta$ prove that $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left[\frac{x+y}{1-xy} \right]$	2
	ii) Hence evaluate $\tan^{-1}(\frac{1}{2}) + \tan^{-1}(\frac{1}{3})$	1

Mark

c) i) Prove that $\frac{d}{dx}(x^2 \tan^{-1} x)$ may be written as

$$2x \tan^{-1} x + 1 - \frac{1}{x^2+1} \quad 2$$

ii) Hence find $\int_0^{\sqrt{3}} x \cdot \tan^{-1} x \, dx$ in exact form 2

$$\therefore \int \sec^2 x \cdot \tan x \, dx$$

$$= \int \sec^2 x \cdot u \cdot \frac{1}{\sec^2 x} \, du$$

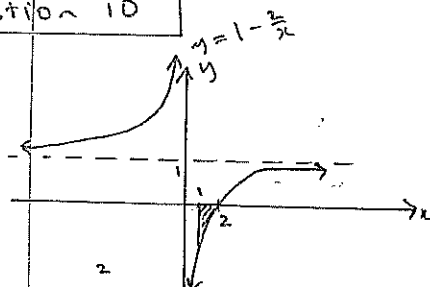
$$= \int u \, du$$

$$= \frac{u^2}{2} + C$$

$$= \frac{\tan^2 x}{2} + C$$

Question 10

a) i)



$$V = \pi \int_1^2 \left(1 - \frac{2}{x}\right)^2 dx$$

$$= \pi \int_1^2 \left(1 - \frac{4}{x} + \frac{4}{x^2}\right) dx$$

$$= \pi \left[x - 4 \ln x - \frac{4}{x} \right]_1^2$$

$$= \pi \left[(2 - 4 \ln 2 - 2) - (1 - 4 \ln 1 - 4) \right]$$

$$= \pi \left[-4 \ln 2 + 3 \right]$$

$$= \pi \left[3 - 4 \ln 2 \right]$$

b) i) $y = \frac{e^x}{x^2 + 1}$

$$u = e^x \quad v = x^2 + 1$$

$$u' = e^x \quad v' = 2x$$

$$\frac{dy}{dx} = \frac{e^x(x^2 + 1) - 2xe^x}{(x^2 + 1)^2}$$

$$= \frac{e^x(x^2 + 1 - 2x)}{(x^2 + 1)^2}$$

st pt if $\frac{dy}{dx} = 0$

$$\therefore x^2 - 2x + 1 = 0$$

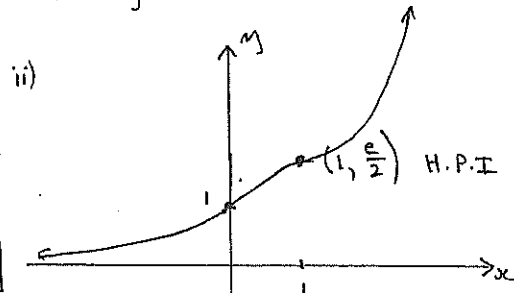
$$(x - 1)^2 = 0$$

$$x = 1$$

test max/min

x	0	1	2
y'	+	0	+

$\therefore$ gradient +ve on either side of $x = 1 \therefore$ H.P.I. on a rising curve.



as $x \rightarrow \infty$ $y \rightarrow \infty$

as $x \rightarrow -\infty$ $y \rightarrow 0$

Question 11

a) i) LHS = $\frac{1}{x-2} - \frac{1}{x+2}$

$$= \frac{(x+2) - (x-2)}{(x-2)(x+2)}$$

$$= \frac{4}{x^2 - 4}$$

$$= \text{RHS}$$

ii) $\int_3^6 \frac{1}{x^2 - 4} dx = \frac{1}{4} \int_3^6 \frac{4}{x^2 - 4} dx$

$$= \frac{1}{4} \int_3^6 \frac{1}{x-2} - \frac{1}{x+2} dx$$

$$= \frac{1}{4} \left[\ln(x-2) - \ln(x+2) \right]_3^6$$

$$= \frac{1}{4} \left[\ln \left(\frac{x-2}{x+2} \right) \right]_3^6$$

$$= \frac{1}{4} \left[\ln \frac{4}{8} - \ln \frac{1}{5} \right]$$

$$= \frac{1}{4} \ln \frac{5}{2}$$

b) i) $\alpha = \tan^{-1} x \quad \beta = \tan^{-1} y$

$$\tan \alpha = x \quad \tan \beta = y$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$= \frac{x + y}{1 - xy}$$

$$\therefore \alpha + \beta = \tan^{-1} \left[\frac{x + y}{1 - xy} \right]$$

$$\therefore \tan^{-1} x + \tan^{-1} y = \tan^{-1} \left[\frac{x + y}{1 - xy} \right]$$

ii) $\therefore \tan^{-1} \left(\frac{1}{2} \right) + \tan^{-1} \left(\frac{1}{3} \right)$

$$= \tan^{-1} \left[\frac{1/2 + 1/3}{1 - 1/2 \cdot 1/3} \right]$$

$$= \tan^{-1} \left[\frac{5/6}{5/6} \right]$$

$$= \tan^{-1} 1$$

$$= \pi/4$$

c) $u = x^2 \quad v = \tan^{-1} x$

i) $u' = 2x \quad v' = \frac{1}{1+x^2}$

$$\therefore \frac{d}{dx} (x^2 \cdot \tan^{-1} x) = 2x \cdot \tan^{-1} x +$$

$$= 2x \cdot \tan^{-1} x + \frac{1+x^2}{1+x^2}$$

$$= 2x \cdot \tan^{-1} x + 1 - \frac{1}{1+x^2}$$

ii) $\left[\frac{d}{dx} (x^2 \cdot \tan^{-1} x) \right] - 1 + \frac{1}{1+x^2} = 2x \tan^{-1} x$

$$\therefore \int_0^{\sqrt{3}} x \tan^{-1} x \, dx = \frac{1}{2} \left[x^2 \cdot \tan^{-1} x \right]_0^{\sqrt{3}} - \int_0^{\sqrt{3}} \left[x - \tan^{-1} x \right] dx$$

$$= \frac{1}{2} \left\{ \left[3 \tan^{-1} \sqrt{3} \right] - \left[x - \tan^{-1} x \right]_0^{\sqrt{3}} \right\}$$

$$= \frac{1}{2} \left\{ 3 \cdot \frac{\pi}{3} - (\sqrt{3} - \tan^{-1} \sqrt{3}) \right\}$$

$$= \frac{1}{2} \left\{ \pi - \sqrt{3} + \frac{\pi}{3} \right\}$$

$$= \frac{2\pi}{3} - \frac{\sqrt{3}}{2}$$

- 1 A
2 C
3 A
4 A
5 C

Question 6

a) i) $\frac{d}{dx}(e^{\sin x}) = \cos x \cdot e^{\sin x}$

ii) $\frac{d}{dx}(\ln(\cos x)) = \frac{-\sin x}{\cos x}$ OR

iii) $\frac{d}{dx}(\sin^{-1} x) = \frac{-\tan x}{\sqrt{1-x^2}}$ OR

b) i) $\cos^{-1}(-\frac{\sqrt{3}}{2})$
 $= \pi - \cos^{-1} \frac{\sqrt{3}}{2}$
 $= \pi - \frac{\pi}{6}$
 $= \frac{5\pi}{6}$

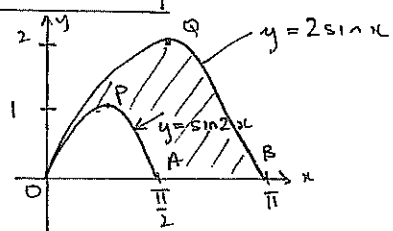
ii) $\tan^{-1}(2 \cos \frac{5\pi}{6})$
 $\cos \frac{5\pi}{6} = -\cos \frac{\pi}{6}$

$= -\sqrt{3}/2$
 $\therefore \tan^{-1}(-\frac{\sqrt{3}}{2})$

$\tan^{-1}(-\sqrt{3})$
 $= -\tan^{-1} \sqrt{3}$
 $= -\pi/3$

$\int_0^1 \frac{1}{\sqrt{4-x^2}} dx = \left[\sin^{-1} \frac{x}{2} \right]_0^1$
 $= \sin^{-1} \frac{1}{2} = \pi/6$

Question 7



$A = \int_0^{\pi/2} 2 \sin x dx - \int_0^{\pi/2} \sin 2x dx$
 $= [-2 \cos x]_0^{\pi/2} - [-\frac{1}{2} \cos 2x]_0^{\pi/2}$
 $= (2 - -2) - (\frac{1}{2} - -\frac{1}{2})$
 $= 3 \text{ unit}^2$

b) $u = x$ $v = \ln x$

i) $u' = 1$ $v' = \frac{1}{x}$

$\therefore \frac{d}{dx}(x \ln x) = \ln x + 1$

ii) $\int_e^{e^2} \frac{1 + \ln x}{x \ln x} dx = \left[\ln(x \ln x) \right]_e^{e^2}$

$= \ln(e^2 \ln e^2) - \ln(e \ln e)$
 $= \ln(e^2 \cdot 2 \ln e) - 1$
 $= \ln 2e^2 - 1$
 $= \ln 2 + \ln e^2 - 1$
 $= \ln 2 + 2 \ln e - 1$
 $= \ln 2 + 1$

c) i) $\cos 2x = \cos^2 x - \sin^2 x$
 $= 1 - 2 \sin^2 x$

ii) $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2} = \lim_{x \rightarrow 0} \frac{1 - (1 - 2 \sin^2 x)}{x^2}$
 $= \lim_{x \rightarrow 0} \frac{2 \sin^2 x}{x^2} = 2$

Question 8

a) $y = a \cdot 10^{bx}$
 $\log_{10} y = \log_{10}(a \cdot 10^{bx})$

$\log_{10} y = \log_{10} a + \log_{10} 10^{bx}$

$\log_{10} y = \log_{10} a + bx \log_{10} 10$

$bx = \log_{10} y - \log_{10} a$

$bx = \log_{10} \left(\frac{y}{a} \right)$

$\therefore x = \frac{1}{b} \log_{10} \left(\frac{y}{a} \right)$

b)

$\sin x = -\frac{1}{2} \therefore x = n\pi + (-1)^n \sin^{-1} \left(-\frac{1}{2} \right)$

$\therefore x = n\pi - (-1)^n \frac{\pi}{6}$

where n is an integer

c) $\frac{dy}{dx} = \frac{2}{x+1}$

$y = 2 \ln(x+1) + c$

sub (0, 1) $\therefore 1 = 2 \ln 1 + c$

$\therefore c = 1$

curve $y = 2 \ln(x+1) + 1$

d) $\sin 2x = 2 \sin x \cdot \cos x$

$\therefore \sin x \cdot \cos x = \frac{1}{2} \sin 2x$

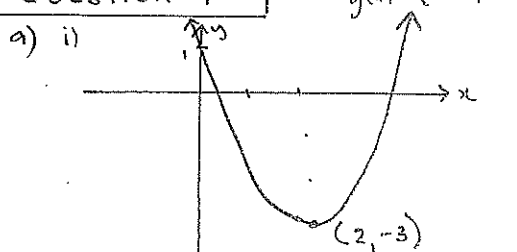
$\sin^2 x \cdot \cos^2 x = \frac{1}{4} \sin^2 2x$

$\therefore \int \sin^2 x \cos^2 x dx = \frac{1}{4} \int \sin^2 2x dx$

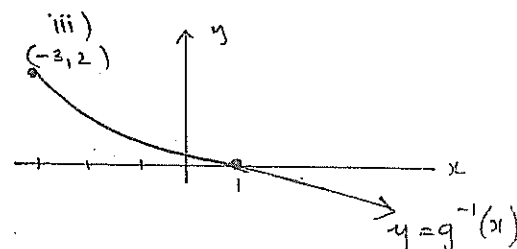
$= \frac{1}{4} \int \frac{1}{2} (1 - \cos 4x) dx$

$= \frac{1}{8} \left[x - \frac{1}{4} \sin 4x \right] + c$

Question 9



ii) D: $x \leq 2$



$x = (y-2)^2 - 3$

$x+3 = (y-2)^2$

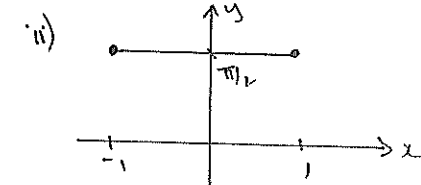
$-\sqrt{x+3} + 2 = y$

$\therefore g^{-1}(x) = -\sqrt{x+3} + 2$

b) i) $\frac{d}{dx}(\cos^{-1} x + \sin^{-1} x)$

$= \frac{-1}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1-x^2}}$

$= 0$



c) $u = \tan x$

$\frac{du}{dx} = \sec^2 x$

$\therefore dx = \frac{1}{\sec^2 x}$